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Validating Authenticity of Electronic Documents Shared via Computer Networks

Abstract

Authentication of electronic document is based on multiple digital signatures incorporated into a blockchain. Structured data, metadata, and instructions may be hashed to generate the multiple digital signatures for distribution via the blockchain. Any peer receiving the blockchain may then verify an authenticity of an electronic document based on any one or more of the multiple digital signatures incorporated into the blockchain.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. application Ser. No. 18/540,677 filed Dec. 14, 2023, which is a continuation of U.S. application Ser. No. 17/322,994 filed May 18, 2021 and since issued as U.S. Pat. No. 11,836,686, which is a continuation of U.S. application Ser. No. 16/548,963 filed Aug. 23, 2019 and since issued as U.S. Pat. No. 11,044,100, which is a continuation of U.S. application Ser. No. 15/419,033 filed Jan. 30, 2017 and since issued as U.S. Pat. No. 10,419,225, with all the above patent applications incorporated herein by reference in their entireties.

BACKGROUND

[0002] Authenticity is important in today's online environment. Electronic documents are easily distributed, especially via private and public networks (such as the Internet). Electronic documents are thus easily altered for fraudulent activities.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0003] The features, aspects, and advantages of the exemplary embodiments are understood when the following Detailed Description is read with reference to the accompanying drawings, wherein:

[0004] FIGS. 1-3 are simplified illustrations of validating an electronic document, according to exemplary embodiments;

[0005] FIGS. 4-7 are detailed illustrations of an operating environment, according to exemplary embodiments;

[0006] FIGS. 8-12 further illustrate verification, according to exemplary embodiments;

[0007] FIG. 13 illustrates multiple digital signatures, according to exemplary embodiments;

[0008] FIG. 14 illustrates a blockchain, according to exemplary embodiments;

[0009] FIG. 15 illustrates multiple verifications, according to exemplary embodiments;

[0010] FIGS. 16-17 are flowcharts illustrating a method of authentication, according to exemplary embodiments;

[0011] FIG. 18 illustrates a fraud alert, according to exemplary embodiments; and

[0012] FIGS. 19-20 depict still more operating environments for additional aspects of the exemplary embodiments.

DETAILED DESCRIPTION

[0013] The exemplary embodiments will now be described more fully hereinafter with reference to the accompanying drawings. The exemplary embodiments may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. These embodiments are provided so that this disclosure will be thorough and complete and will fully convey the exemplary embodiments to those of ordinary skill in the art. Moreover, all statements

herein reciting embodiments, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future (i.e., any elements developed that perform the same function, regardless of structure).

[0014] Thus, for example, it will be appreciated by those of ordinary skill in the art that the diagrams, schematics, illustrations, and the like represent conceptual views or processes illustrating the exemplary embodiments. The functions of the various elements shown in the figures may be provided through the use of dedicated hardware as well as hardware capable of executing associated software. Those of ordinary skill in the art further understand that the exemplary hardware, software, processes, methods, and/or operating systems described herein are for illustrative purposes and, thus, are not intended to be limited to any particular named manufacturer.

[0015] As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless expressly stated otherwise. It will be further understood that the terms “includes,” “comprises,” “including,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. Furthermore, “connected” or “coupled” as used herein may include wirelessly connected or coupled. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0016] It will also be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first device could be termed a second device, and, similarly, a second device could be termed a first device without departing from the teachings of the disclosure.

[0017] FIGS. **1-3** are simplified illustrations of validating an electronic document **20**, according to exemplary embodiments. Here exemplary embodiments verify whether the electronic document **20** has been altered since creation. As the reader may understand, an authenticity **22** of the electronic document **20** may be very important in banking, security, identification, and other activities conducted via the Internet. If the electronic document **20** has changed since its original creation, then banking and other online activities may be compromised. Indeed, an altered version **24** of the electronic document **20** may indicate malicious or fraudulent activity is being attempted. Exemplary embodiments may thus include a security scheme based on electronic data **26** representing the electronic document **20**.

[0018] FIG. **1** illustrates a verification server **28**. The verification server **28** determines whether the electronic document **20** has been changed since creation **30**. The verification server **28** obtains the electronic data **26** representing an original version **32** of the electronic document **20**. This disclosure mostly explains the original version **32** as of a date **34** of creation. The original version **32**, though, may be demarcated with reference to any time, date, or version. Regardless, the verification server **28** may generate one or more hash values **36** associated with a hash tree **38** based on the electronic data **26** representing the original version **32** of the electronic document **20**. The hash tree **38** may be generated using an electronic representation of a hashing algorithm **40**. The verification server **28** may then distribute one or more of the hash values **36** via a blockchain **50**. That is, any of the hash values **36** (e.g., hash list, hash chain, branches, nodal leaves) may be added to, stored in, or incorporated in, any record, transaction, or block and distributed via the blockchain **50**. While the blockchain **50** may be sent or routed to any destination (such as an Internet Protocol address associated with another server), FIG. **1** illustrates peer distribution. That is, the verification server **28** may broadcast the blockchain **50** to the IP addresses associated with a network **52** of peer devices or nodes for verification. Exemplary embodiments may thus utilize the

blockchain 50 as a distribution or publication mechanism. As the reader may understand, the blockchain 50 is generally a digital ledger in which transactions are chronologically and/or publically recorded. The blockchain 50 is most commonly used in decentralized cryptocurrencies (such as Bitcoin). The blockchain 50, however, may be adapted to any chain or custody (such as in medical records and in chains of title in real estate transactions). Indeed, there are many different mechanisms and configurations of the blockchain 50, and exemplary embodiments may be adapted to any version. Because peer-to-peer blockchain technology is generally known, this disclosure need not provide a detailed explanation.

[0019] FIG. 2 illustrates a verification scheme. Here the verification server 28 determines whether a current version 60 of the electronic document 20 has changed since the date 34 of creation. That is, the verification server 28 determines whether the current version 60 differs from the original version 32. The verification server 28 retrieves electronic data 61 representing the current version 60 of the electronic document 20. The verification server 28 may then generate one or more verification hash values 62 based on the electronic data 61 representing the current version 60. The verification hash values 62 are generated using the electronic representation of the hashing algorithm 40. The verification server 28 may then compare the verification hash values 62 to the hash values 36 representing the original version 32 (as of the date 34 of creation). If the verification hash values 62 satisfactorily compare to the hash values 36, then the verification server 28 may infer or determine that the current version 60 is the same as the original version 32. If, however, the verification hash values 62 fail to satisfy the hash values 36, then the verification server 28 may infer that the current version 60 differs from the original version 32. The current version 60 of the electronic document 20, in other words, has been altered since the date 34 of creation. The verification server 28 may thus generate a fraud alert 64 to implement enhanced security measures.

[0020] Exemplary embodiments may detect minor changes. The hashing algorithm 40 is very sensitive to even subtle alterations in the electronic document 20. There are many hashing algorithms, and exemplary embodiments may utilize any of the hashing algorithms. For simplicity, though, this disclosure will mostly discuss the SHA family of cryptographic hashing algorithms, which many readers are thought familiar. For example, if the SHA-256 algorithm is applied to the electronic data 26 representing the original version 32, the result is a 256-bit digital signature. There is thus an extremely low probability that different source data would produce the same digital signature. So, if the current version 60 of the electronic document 20 has even a subtle change (such as a single character in a single textual word or number), its corresponding digital signature (e.g., the verification hash values 62) will not match or equal the hash values 36 representing the original version 32. Exemplary embodiments thus quickly determine whether the electronic document 20 has been changed since the date 34 of creation.

[0021] FIG. 3 illustrates nodal verification. Here exemplary embodiments may distribute one or more of the hash values 36 via the blockchain 50. Once the verification server 28 generates the hash values 36 (based on the electronic data 26 representing the original version 32 of the electronic document 20), the verification server 28 may distribute the hash values 36 via the blockchain 50. For example, the verification server 28 may incorporate the 256-bit digital signature 70 (generated by the SHA-256 algorithm) into the blockchain 50. Any trusted member of the network 52 of peer devices may then verify the current version 60 of the electronic document 20. A trusted peer device 72, for example, may receive the blockchain 50 and retrieve or determine the digital signature 70 incorporated therein. The trusted peer device 72 may then itself retrieve the electronic data 71 representing the current version 60 and generate the verification hash values 62 (such as a 256-bit verification digital signature 74). The trusted peer device 72 may then compare the hash values 36 (received via the blockchain 50) to the verification hash values 62 generated based on the current version 60 of the electronic document 20. As a simple example, the trusted peer device 72 may compare the 256-bit digital signature 70 (based on the original version 32) to the 256-bit verification digital signature 74 (based on the current version 60). If the verification

hash values **36** (such as a 256-bit verification digital signature **74**) satisfactorily compare to the hash values **36** (such as the 256-bit digital signature **70**) received via the blockchain **50**, then the trusted peer device **72** may infer or determine that the current version **60** is authentic and unaltered. However, if the verification hash values **36** (e.g., the 256-bit verification digital signature **74**) fails to satisfy the hash values **36** (e.g., the 256-bit digital signature **70**), then the trusted peer device **72** may infer that the current version **60** is inauthentic and altered. The trusted peer device **72** may thus generate the fraud alert **64** to implement enhanced security measures.

[0022] FIGS. **4-7** are detailed illustrations of an operating environment, according to exemplary embodiments. FIG. **4** illustrates the verification server **28** communicating with the trusted peer device **72** via a communications network **80** and/or via a wireless network **82**. FIG. **4** illustrates the trusted peer device **72** as a mobile smartphone **84**, which most readers are thought familiar. The trusted peer device **72**, though, may be any processor-controlled device, as later paragraphs will explain. The verification server **28** may have a processor **86** (e.g., “ μ P”), application specific integrated circuit (ASIC), or other component that executes a verification algorithm **86** stored in a local memory device **88**. The verification algorithm **86** includes instructions, code, and/or programs that cause the verification server **28** to perform operations, such as generating the hash values **36** associated with the electronic document **20** (as the above paragraphs explained). The verification algorithm **86** may thus generate the digital signature **70** representing the original version **32** (using the hashing algorithm **40**). The verification algorithm **86** may then instruct or cause the verification server **28** to integrate the digital signature **70** into the blockchain **50** for distribution to the mobile smartphone **84**. Exemplary embodiments, though, may send the digital signature **70** and/or the blockchain **50** to any IP address associated with any network destination or device.

[0023] FIG. **5** illustrates peer verification. Now that the blockchain **50** is distributed, the trusted peer device **72** (again illustrated as the mobile smartphone **84**) may determine the authenticity **22** of the current version **60** of the electronic document **20**. FIG. **5** thus illustrates the mobile smartphone **84** receiving the blockchain **50** and the digital signature **70** integrated therein. The mobile smartphone **84** has a processor **90**, application specific integrated circuit (ASIC), or other component that executes a peer-side verification algorithm **92** stored in a local memory device **94**. The peer-side verification algorithm **92** includes instructions, code, and/or programs that cause the processor **90** to perform operations, such as verifying the current version **60** of the electronic document **20**. The peer-side verification algorithm **92** causes the mobile smartphone **84** to retrieve the electronic data **71** representing the current version **60** and to generate the verification digital signature **74** (such as the corresponding 256-bit hash). If the verification digital signature **74** satisfactorily compare to the digital signature **70** received via the blockchain **50**, then the peer-side verification algorithm **92** may infer or determine that the current version **60** is authentic and unaltered. However, if the verification digital signature **74** fails to satisfy the digital signature **70** received via the blockchain **50**, then the peer-side verification algorithm **92** may infer that the current version **60** is inauthentic and altered. The peer-side verification algorithm **92** may thus generate the fraud alert **64** to implement enhanced security measures.

[0024] FIG. **6** illustrates servile verification. Here the verification server **28** may itself determine the authenticity **22** of different versions of the electronic document **20**. Whenever the verification server **28** receives the current version **60** of the electronic document **20**, the verification server **28** may alert or notify of security concerns. Here the verification algorithm **86** causes the verification server **28** to retrieve the current version **60** and to generate the corresponding verification digital signature **74**. If the verification digital signature **74** satisfactorily compares to the digital signature **70** (representing the original version **32**), then the verification algorithm **86** may infer or determine that the current version **60** is authentic and unaltered. However, if the verification digital signature **74** fails to satisfy the digital signature **70** (representing the original version **32**), then the verification algorithm **86** may implement the fraud alert **64**, as the current version **60** is inauthentic and/or altered.

[0025] FIG. 7 illustrates the general verification scheme. Again, because many readers may be familiar with the SHA-256 hashing algorithm, the general verification scheme may use the 256-bit hash value computed by the SHA-256 algorithm. Exemplary embodiments obtain or retrieve the electronic data **26** representing the original version **32** of the electronic document **20**. The electronic data **26** is acted on by the SHA-256 hashing algorithm to generate a 256-bit hash value as the digital signature **70**. Exemplary embodiments may also obtain or retrieve the electronic data **71** representing the current version **60** of the electronic document **20**. The electronic data **71** is acted on by the SHA-256 hashing algorithm to generate a corresponding 256-bit hash value as the verification digital signature **74**. If a match is determined, exemplary embodiments may infer that the current version **60** is authentic and unaltered. However, if the digital signatures **70** and **74** fail to match, exemplary embodiments may infer that the current version **60** is inauthentic and generate the fraud alert **64**.

[0026] Exemplary embodiments may be applied regardless of networking environment. Exemplary embodiments may be easily adapted to stationary or mobile devices having cellular, wireless fidelity (WI-FI®), near field, and/or BLUETOOTH® capability. Exemplary embodiments may be applied to mobile devices utilizing any portion of the electromagnetic spectrum and any signaling standard (such as the IEEE 802 family of standards, GSM/CDMA/TDMA or any cellular standard, and/or the ISM band). Exemplary embodiments, however, may be applied to any processor-controlled device operating in the radio-frequency domain and/or the Internet Protocol (IP) domain. Exemplary embodiments may be applied to any processor-controlled device utilizing a distributed computing network, such as the Internet (sometimes alternatively known as the “World Wide Web”), an intranet, a local-area network (LAN), and/or a wide-area network (WAN). Exemplary embodiments may be applied to any processor-controlled device utilizing power line technologies, in which signals are communicated via electrical wiring. Indeed, exemplary embodiments may be applied regardless of physical componentry, physical configuration, or communications standard(s).

[0027] Exemplary embodiments may utilize any processing component, configuration, or system. Any processor could be multiple processors, which could include distributed processors or parallel processors in a single machine or multiple machines. The processor can be used in supporting a virtual processing environment. The processor could include a state machine, application specific integrated circuit (ASIC), programmable gate array (PGA) including a Field PGA, or state machine. When any of the processors execute instructions to perform “operations”, this could include the processor performing the operations directly and/or facilitating, directing, or cooperating with another device or component to perform the operations.

[0028] Exemplary embodiments may packetize. The verification server **28** and the trusted peer device **72** may have network interfaces to the communications network **80**, thus allowing collection and retrieval of information. The information may be received as packets of data according to a packet protocol (such as the Internet Protocol). The packets of data contain bits or bytes of data describing the contents, or payload, of a message. A header of each packet of data may contain routing information identifying an origination address and/or a destination address associated with any of the verification server **28** and the trusted peer device **72**.

[0029] FIGS. 8-12 further illustrate verification, according to exemplary embodiments. Here exemplary embodiments authenticate the electronic document **20**, based on its electronic data **26**. While exemplary embodiments are applicable to any type of the electronic document **20**, most readers are thought familiar with a driver's license **100**. That is, the electronic document **20** may have content representing an electronic version of the driver's license **100**. While the electronic document **20** may be formatted according to any format **102**, most readers are thought familiar with a portable document format (“PDF”) **104**, the MICROSOFT® WORD® extensible markup language extension (“docx”) **106**, and/or the extensible markup language (“XML”) **108**. Exemplary embodiments may thus retrieve the electronic data **26** representing the driver's license **100** and

generate the corresponding digital signature **70** (such as the corresponding 256-bit hash if using the SHA-256 hashing algorithm **40**). Exemplary embodiments may then incorporate the digital signature **70** into the blockchain **50** for distribution via the communications network **80**. Any destination (such as the trusted peer device **72**) may thus verify an alleged copy of the driver's license **100** against the blockchain **50** (as this disclosure above explains).

[0030] Exemplary embodiments may be applied to any file formatting and/or specification. The format **102** may be proprietary, free, unpublished, and/or open. The format **102** may be designed for images, containers, audio, video, text, subtitles, control characters, and encoding schemes. The format **102** may be HTML, vector graphics, source code, text files, syntax, and software programming.

[0031] FIG. **9** illustrates structured data **110**. As the reader may understand, the electronic data **26** representing the electronic document **20** may be the structured data **110**. That is, the structured data **110** may be organized (such as an entry **112** or database field **114** in a relational spreadsheet **116** or database **118**), contained within a fixed data field **120** or data record **122**, and/or be addressable via a network or memory address **124**. Again referencing the electronic version of the driver's license **100**, the structured data **110** may be organized according to the JavaScript Object Notation (or "JSON"). As the JavaScript Object Notation is a known format for structuring data, the JSON format need not be explained in detail. Suffice it to say that at least some of the electronic data **26** representing the driver's license **100** may be a JSON document **126** having the structured data **110** arranged as fields [{"Name": "Bob Smith"}, {"State": "TX"}, {"Birth": "May 1, 1975"} . . .]. The driver's license **100** may thus be formatted according to a JSON schema **128** and stored in an electronic database **130** (perhaps having other structured data **110**, such as electronic database associations according to [{"Title": "Drivers License", "properties": {"Name": {"type": "string"} . . . }].

[0032] Exemplary embodiment may thus incorporate a data version **132** in the blockchain **50**. For example, if the driver's license **100** is the JSON document **126**, then the data version **132** may be the structured data **110** arranged or formatted according to the JSON schema **128**. Exemplary embodiments may thus retrieve the data version **132** and generate the corresponding digital signature **70** (such as the 256-bit hash value using the SHA-256 hashing algorithm **40**). Exemplary embodiments may then incorporate the digital signature **70** into the blockchain **50** for distribution (such as to the trusted peer device **72**). The trusted peer device **72** may thus verify an alleged copy of the driver's license **100** against the blockchain **50** (as this disclosure above explains). Moreover, once the structured data **110** is known (such as JSON schema **128**), any electronic document referenced in the electronic database **130** may be recreated, hashed, and checked against the blockchain **50** to ensure the electronic data **26** has not been altered. For example, if the electronic data **26** representing the driver's license **100** is stored in a human resources database, then exemplary embodiments permit recreating the driver's license **100** (perhaps via a POSTGRES® database) and authentication.

[0033] FIGS. **10-11** illustrate metadata **140**. Here the electronic data **26** representing the electronic document **20** (such as the driver's license **100**) may include the metadata **140** (such as [{"Creation Time": "2012-05-07T11:12:32"}, {"SourceID": "1131122"}, {"Location": "Wells Fargo System XXX, ID YYY"} . . .]). Exemplary embodiments may thus retrieve the metadata **140** and generate the corresponding digital signature **70** (such as the 256-bit hash value representing the metadata **140**). Exemplary embodiments may then incorporate the digital signature **70** representing the metadata **140** into the blockchain **50** for distribution to the trusted peer device **72**. The trusted peer device **72** may thus perform a two-fold verification. That is, any alteration to the driver's license **100** may be determined (as this disclosure explains), but exemplary embodiments may also verify that the date **34** of creation (e.g., [{"Creation Time": "2012-05-07T11:12:32"}]) has not changed. Again, if the digital signature **70** representing the metadata **140** has changed over time (such as when comparing the original version **32** to the current version **60**), then exemplary embodiments

may flag or notify of a possible fraud attempt. Any change in the digital signature **70** over time may also be useful for audit trails in banking, mortgage processing, and other activities.

[0034] FIG. **11** illustrates different types of the metadata **140**. While exemplary embodiments may hash any type of the metadata **140**, this disclosure will mainly describe the metadata **140** thought familiar to most readers. For example, the metadata **140** may describe the date **112** of creation, a title, an author, a location (such as GPS information at creation), word/character count, and an abstract describing or summarizing the electronic document **20**. The metadata **140** may also include one or more keywords associated with the electronic document **20**. The metadata **140** may also include a file hierarchy where the electronic document **20** is stored and/or a network address for retrieval. The network address, for example, may be associated with a server or other machine locally or remotely storing the electronic document **20**. The metadata **140** may also include structural details, such as file size, page numbering, chapter organization, and image data. Other metadata **140** may describe approved users (such as administrator and user permissions or identities) and digital rights management (or “DRM”). The metadata **140** may be formatted according to any standard.

[0035] FIG. **12** illustrates instructions **150**. Here the electronic data **26** representing the driver's license **100** may include the instructions **150**. While exemplary embodiments may be applicable to any instructions, the instructions **150** may be structured (such as executable code), unstructured instructions (such as non-executable commentary lines in code, such as English language “do thing 1, then thing 2, then thing 3”). Other instructions **150** may include any messages (such as “When this document is accessed, POST to the URL <http://some.target.url>”). Exemplary embodiments may thus retrieve the instructions **150**, generate the corresponding digital signature **70** (such as the 256-bit hash value representing the instructions **150**), and incorporate into the blockchain **50**. Again, if the digital signature **70** representing the instructions **150** has changed over time, then exemplary embodiments may flag or notify of a possible fraud attempt.

[0036] FIG. **13** illustrates multiple digital signatures **160**, according to exemplary embodiments. Because the electronic document **20** may comprise or contain multiple types of the electronic data **26**, exemplary embodiments may generate the multiple digital signatures **160**. Exemplary embodiments, for example, may generate a first digital signature **70a** (such as the 256-bit hash value using the SHA-256 hashing algorithm **40**) based on the data version **132** associated with the electronic document **20** (as above explained with reference to FIGS. **8-9**). Exemplary embodiments may generate a second digital signature **70b** based on the metadata **140** contained within, or associated with, the electronic document **20** (as above explained with reference to FIGS. **10-11**). Exemplary embodiments may generate a third digital signature **70c** based on the instructions **150** contained within, or associated with, the electronic document **20** (as above explained with reference to FIG. **12**). Any or all of the multiple digital signatures **160** may be generated based on the electronic data **26** representing the electronic document **20**.

[0037] FIG. **14** further illustrates the blockchain **50**, according to exemplary embodiments. Once any of the multiple digital signatures **160** is/are generated, exemplary embodiments may incorporate one or more of the multiple digital signatures **160** into the blockchain **50**. That is, any one or more of the multiple digital signatures **160** may be added to, stored in, or incorporated into any record, transaction, or block within the blockchain **50**. FIG. **14**, for simplicity, illustrates the blockchain **50** including the first digital signature **70a** (based on the data version **132**), the second digital signature **70b** (based on the metadata **140**), and the third digital signature **70c** (based on the instructions **150**). The verification server **28** may then distribute or broadcast the blockchain **50** to the trusted peer device **72** for subsequent verification.

[0038] FIG. **15** illustrates multiple verifications, according to exemplary embodiments. Once any of the multiple digital signatures **160** (e.g., **70a**, **70b**, and/or **70c**) are received (perhaps via the blockchain **50**), the trusted peer device **72** may then use any one or more of the multiple digital signatures **160** to verify the authenticity **22** of the current version **60** of the electronic document **20**.

The peer-side verification algorithm **92**, for example, may cause the trusted peer device **72** to generate a first verification digital signature (“1.sup.st VDS”) **74a** by hashing the data version **132** associated with the current version **60**. The peer-side verification algorithm **92** additionally or alternatively cause the trusted peer device **72** to generate a second verification digital signature (“2.sup.nd VDS”) **74b** by hashing the metadata **140** associated with the current version **60**. The peer-side verification algorithm **92** additionally or alternatively cause the trusted peer device **72** to generate a third verification digital signature (“3.sup.rd VDS”) **74b** by hashing the metadata **140** associated with the current version **60**. If any one or more of the verification digital signatures **74a**, **74b**, and/or **74c** satisfactorily compares to their corresponding digital signatures **70a**, **70b**, and/or **70c**, then the peer-side verification algorithm **92** may infer or determine that the current version **60** is authentic and unaltered. However, if any one or more of the verification digital signatures **74a**, **74b**, and/or **74c** fails to satisfy their corresponding digital signatures **70a**, **70b**, and/or **70c**, then the peer-side verification algorithm **92** may infer that the current version **60** is inauthentic and altered. The peer-side verification algorithm **92** may thus generate the fraud alert **64** to implement enhanced security measures.

[0039] FIGS. **16-17** are flowcharts illustrating a method of authentication, according to exemplary embodiments. The electronic data **26** representing the original version **32** of the electronic document **20** is retrieved (Block **200**) and hashed using the hashing algorithm **40** (Block **202**). One or more of the multiple digital signatures **160** are generated (Block **204**) and incorporated into the blockchain **50** (Block **206**). The blockchain **50** is distributed via the Internet (Block **208**). When the blockchain **50** is received (by any recipient, such as the trusted peer device **72**) (Block **210**), any of the multiple digital signatures **160** may be used verify the current version **60** of the electronic document **20** (Block **212**). The multiple verification digital signatures **74a**, **74b**, and/or **74c** are generated based on the current version **60** of the electronic document **20** (Block **214**).

[0040] The flowchart continues with FIG. **17**. The multiple verification digital signatures **74a**, **74b**, and/or **74c** are compared to the multiple digital signatures **160** incorporated into the blockchain **50** (Block **218**). The number of matches is summed (Block **220**) and compared to a verification threshold (Block **222**). If the number of matches equals or exceeds the verification threshold (Block **224**), then the current version **60** is authentic (Block **226**). However, if the number of matches is less than the verification threshold (Block **224**), then the current version **60** is altered (Block **228**) and fraud alert **64** is generated (Block **230**).

[0041] FIG. **18** illustrates the fraud alert **64**, according to exemplary embodiments. The verification server **28** and/or the trusted peer device **72** generates the fraud alert **64**. While the fraud alert **64** may have any mechanism and structure, FIG. **17** illustrates a simple notification example. The fraud alert **64** is an electronic message **240** that is sent to one or more notification addresses **242**. The electronic message **240** routes via the communications network **80** to a network address (e.g., IP address) associated with a destination device **244**. The electronic message **240** contains information or data describing an inauthenticity of the current version **60** of the electronic document **20**, based on one or more of the non-matching digital signatures **70**.

[0042] FIG. **19** is a schematic illustrating still more exemplary embodiments. FIG. **19** is a more detailed diagram illustrating a processor-controlled device **250**. As earlier paragraphs explained, the verification algorithm **86** and the peer-side verification algorithm **92** may partially or entirely operate in any mobile or stationary processor-controlled device. FIG. **19**, then, illustrates the verification algorithm **86** and the peer-side verification algorithm **92** stored in a memory subsystem of the processor-controlled device **250**. One or more processors communicate with the memory subsystem and execute either, some, or all applications. Because the processor-controlled device **250** is well known to those of ordinary skill in the art, no further explanation is needed.

[0043] FIG. **20** depicts other possible operating environments for additional aspects of the exemplary embodiments. FIG. **20** illustrates the verification algorithm **86** and the peer-side verification algorithm **92** operating within various other processor-controlled devices **250**. FIG. **20**,

for example, illustrates that the verification algorithm **86** and the peer-side verification algorithm **92** may entirely or partially operate within a set-top box (“STB”) (**252**), a personal/digital video recorder (PVR/DVR) **254**, a Global Positioning System (GPS) device **256**, an interactive television **258**, a tablet computer **260**, or any computer system, communications device, or processor-controlled device utilizing any of the processors above described and/or a digital signal processor (DP/DSP) **262**. Moreover, the processor-controlled device **250** may also include wearable devices (such as watches), radios, vehicle electronics, clocks, printers, gateways, mobile/implantable medical devices, and other apparatuses and systems. Because the architecture and operating principles of the various devices **250** are well known, the hardware and software componentry of the various devices **250** are not further shown and described.

[0044] Exemplary embodiments may be applied to any signaling standard. Most readers are thought familiar with the Global System for Mobile (GSM) communications signaling standard. Those of ordinary skill in the art, however, also recognize that exemplary embodiments are equally applicable to any communications device utilizing the Time Division Multiple Access signaling standard, the Code Division Multiple Access signaling standard, the “dual-mode” GSM-ANSI Interoperability Team (GAIT) signaling standard, or any variant of the GSM/CDMA/TDMA signaling standard. Exemplary embodiments may also be applied to other standards, such as the I.E.E.E. 802 family of standards, the Industrial, Scientific, and Medical band of the electromagnetic spectrum, BLUETOOTH®, and any other.

[0045] Exemplary embodiments may be physically embodied on or in a computer-readable storage medium. This computer-readable medium, for example, may include CD-ROM, DVD, tape, cassette, floppy disk, optical disk, memory card, memory drive, and large-capacity disks. This computer-readable medium, or media, could be distributed to end-subscribers, licensees, and assignees. A computer program product comprises processor-executable instructions for verifying authenticity of electronic documents, as the above paragraphs explained.

[0046] While the exemplary embodiments have been described with respect to various features, aspects, and embodiments, those skilled and unskilled in the art will recognize the exemplary embodiments are not so limited. Other variations, modifications, and alternative embodiments may be made without departing from the spirit and scope of the exemplary embodiments.

Claims

1-20. (canceled)

21. A method performed by a server that verifies an authenticity of an electronic document shared via a computer network among computers, the method comprising: receiving, by the server, an electronic data comprising a current version of the electronic document and a metadata associated with the current version of the electronic document; generating, by the server, a current hash value by hashing only the metadata associated with the current version of the electronic document; retrieving, by the server, from the blockchain, a creational hash value, the creational hash value representing a metadata associated with an original version of the electronic document; comparing, by the server, the creational hash value to the current hash value; in response to the current hash value failing to match the creational hash value, determining, by the server, that the current version of the electronic document shared via the computer network among the computers is not an authentic copy of the original version of the electronic document; and generating, by the server, a fraud alert indicating the current version of the electronic document is inauthentic.

22. The method of claim 21, further comprising retrieving a hashing algorithm.

23. The method of claim 22, wherein the hashing algorithm is a SHA-256 algorithm.

24. The method of claim 21, wherein the metadata associated with the original version of the electronic document comprises a structural detail of the original version of the electronic document.

25. The method of claim 21, wherein the metadata associated with the original version of the

electronic document comprises a date of creation of the original version of the electronic document.

26. The method of claim 21, further comprising recording a result of the comparing step on the blockchain.

27. The method of claim 26, wherein the result of the comparing step comprises information describing an inauthenticity of the current version of the electronic document.

28. A system including a hardware processor and a memory device storing instructions that, when executed by the hardware processor, perform operations, the operations comprising: receiving, by the server, an electronic data comprising a current version of the electronic document and a metadata associated with the current version of the electronic document; generating, by the server, a current hash value by hashing only the metadata associated with the current version of the electronic document; retrieving, by the server, from the blockchain, a creational hash value, the creational hash value representing a metadata associated with an original version of the electronic document; comparing, by the server, the creational hash value to the current hash value; in response to the current hash value failing to match the creational hash value, determining, by the server, that the current version of the electronic document shared via the computer network among the computers is not an authentic copy of the original version of the electronic document; and generating, by the server, a fraud alert indicating the current version of the electronic document is inauthentic.

29. The system of claim 28, wherein the operations further comprise retrieving a hashing algorithm.

30. The system of claim 29, wherein the hashing algorithm is a SHA-256 algorithm.

31. The system of claim 28, wherein the metadata associated with the original version of the electronic document comprises a structural detail of the original version of the electronic document.

32. The system of claim 28, wherein the metadata associated with the original version of the electronic document comprises a date of creation of the original version of the electronic document.

33. The system of claim 28, wherein the operations further comprise recording a result of the comparing step on the blockchain.

34. The system of claim 33, wherein the result of the comparing step comprises information describing an inauthenticity of the current version of the electronic document.

35. A memory device storing instructions that when executed by a hardware processor perform operations, the operations comprising: receiving, by the server, an electronic data comprising a current version of the electronic document and a metadata associated with the current version of the electronic document; generating, by the server, a current hash value by hashing only the metadata associated with the current version of the electronic document; retrieving, by the server, from the blockchain, a creational hash value, the creational hash value representing a metadata associated with an original version of the electronic document; comparing, by the server, the creational hash value to the current hash value; in response to the current hash value failing to match the creational hash value, determining, by the server, that the current version of the electronic document shared via the computer network among the computers is not an authentic copy of the original version of the electronic document; and generating, by the server, a fraud alert indicating the current version of the electronic document is inauthentic.

36. The memory device of claim 35, wherein the operations further comprise retrieving a hashing algorithm.

37. The memory device of claim 36, wherein the hashing algorithm is a SHA-256 algorithm.

38. The memory device of claim 35, wherein the metadata associated with the original version of the electronic document comprises a structural detail of the original version of the electronic document.

39. The memory device of claim 35, wherein the metadata associated with the original version of the electronic document comprises a date of creation of the original version of the electronic document.

40. The memory device of claim 35, wherein the operations further comprise recording a result of the comparing step on the blockchain.
